

MIET2572 – Wearable Systems Data Lab

Week 3: Motion as Signal

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Objective

In this lab you will transform raw accelerometer data into a meaningful metric: step count.

Key idea: A wearable does not measure steps. It measures signals and infers them.

Pipeline Overview

Sensor → Signal → Processing → Interpretation

Data Source Selection

This lab can be executed using two types of data:

Option A – Synthetic Data

- Generated signal with controlled properties
- Clean, stable, and predictable
- Useful to understand the behaviour of the algorithm

Option B – Measured Data (Phyphox)

- Real accelerometer measurements
- Includes noise, variability, and artefacts
- Represents real-world conditions

Interpretation

- Synthetic data helps you understand the model
- Real data tests whether the model actually works

Understanding a model in ideal conditions is not the same as validating it in real conditions.

Task 1 – Walking Data

A model is understood under control, but validated under reality.

Action

Step 1 – Controlled Condition

- Run the pipeline using synthetic data (controlled condition)
- (Hint: toggle between synthetic and measured data in the notebook)

Step 2 – Measured Condition

- Record 30 s of walking using Phyphox (100 Hz)
- Export the CSV file
- Upload the data into the notebook
- Run the same pipeline using your measured data

Analyse

- Plot raw signal
- Compute magnitude:

$$a_{mag} = \sqrt{a_x^2 + a_y^2 + a_z^2}$$

- Apply smoothing
- Detect peaks

Compare

- Manual step count vs algorithm output

Question

- Do both counts match? Why or why not?

Task 2 – Modify the Model

A model is defined by its parameters.

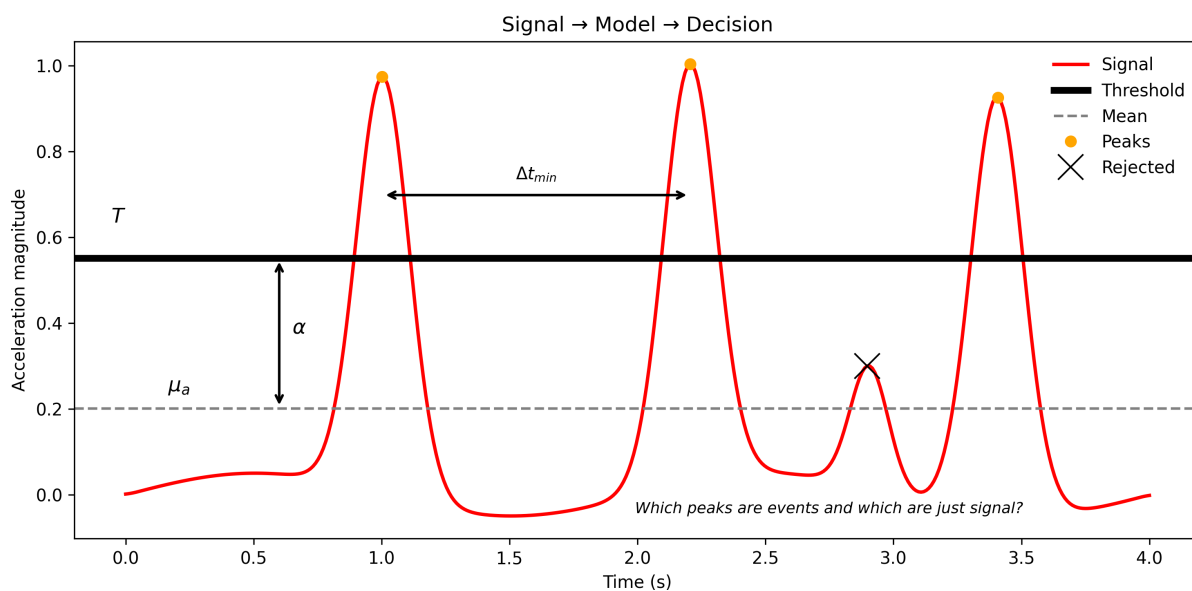


Figure 1: Peak detection parameters. The threshold $T = \mu_a + \alpha$ defines which peaks are valid. The minimum time Δt_{min} prevents multiple detections of the same step.

The algorithm does not detect steps. It detects peaks under constraints.

Action

Step 1 – Threshold Variation

- Modify the amplitude threshold parameter α
- Run the pipeline and record the step count

Model–Code Mapping

- Threshold: $T = \mu_a + \alpha$
- Minimum time between peaks: $\Delta t_{min} = 0.4$ s

Where

- T is the detection threshold
- μ_a is the mean value of the acceleration magnitude signal
- α is the amplitude offset above the baseline
- Δt_{min} is the minimum time allowed between consecutive detected peaks

Peak Detection Rule

- Local maximum
- Above threshold T
- Separated by at least Δt_{min}

Analyse

- How does step count change when α increases?
- How does smoothing affect peak detection?

Question

- Which parameter has a stronger effect?

If small parameter changes produce large output differences, is your system reliable?

Action

Step 2 – Smoothing Variation

- How does smoothing change what the algorithm sees?
- Modify the smoothing window size
- Run the pipeline and record the step count

Smoothing Parameter

Before peak detection, the signal is smoothed using a moving average:

- **Window size:** $N = 0.2 \cdot f_s$

Where

- N is the number of samples in the smoothing window
- f_s is the sampling frequency

Interpretation

- A larger window produces a smoother signal but may remove peaks
- A smaller window preserves detail but retains noise

Smoothing does not remove noise. It reshapes the signal before detection.

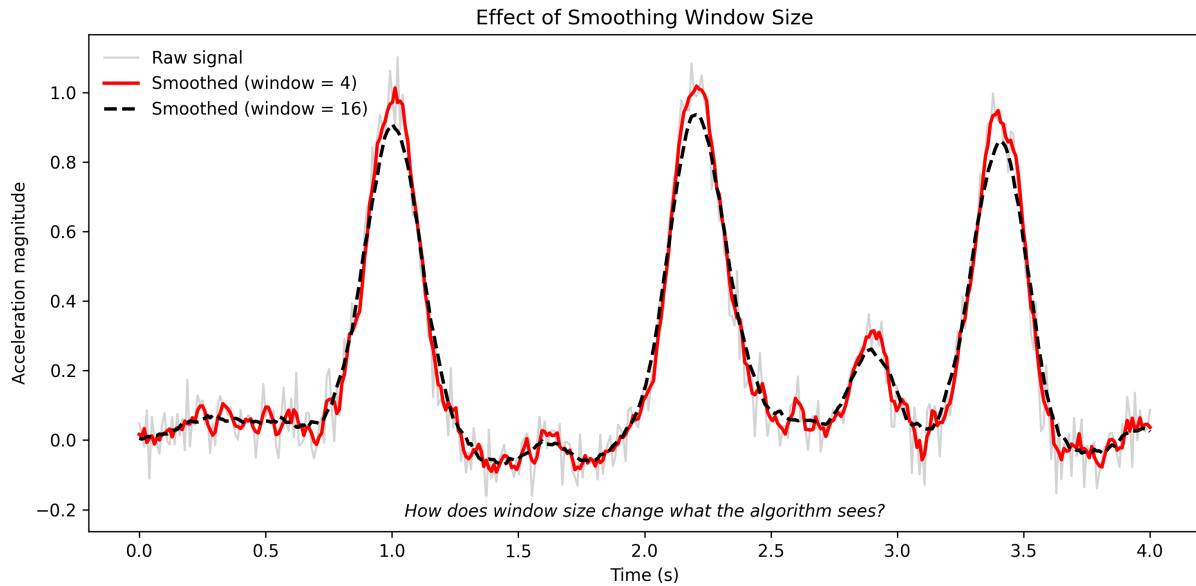


Figure 2: Smoothing changes what the algorithm sees. The window size controls the balance between noise reduction and signal detail, directly influencing which peaks survive to be detected.

Results Log

Condition	α	Window Size	Step Count
Walking (baseline)			
Walking (modified)			
Jumping			

Task 3 – Jumping Data

A model built for one behaviour may fail in another.

Does the model detect steps, or just peaks?

Action

Step 1 – Load Data

- Load data from jumping activity.

Step 2 – Apply Pipeline

- Apply the same processing pipeline used for walking

Analyse

- Compare signal shape with walking data
- Observe peak amplitude and temporal spacing

Question

- Does the step detection model still behave correctly?
- Which assumption is no longer valid?

Task 4 – Model Assumptions

A model works only under certain assumptions.

Model Reminder

- A peak must exceed a threshold T
- Peaks must be separated by at least Δt_{min}

Interpretation

- The model assumes periodic motion
- The model assumes consistent amplitude
- The model assumes regular timing between events

Question

- What defines a “step” in this model?
- What assumptions are required for it to work?

Write

- Define a “step” in terms of signal properties (amplitude, timing, threshold)

Task 5 – Model Failure Across Scenarios

Action

- Compare the results for walking and jumping

Analyse

- How does the signal change between scenarios?
- Does the model still detect “steps” correctly?

Question

- Which assumptions identified in Task 4 fail in the jumping case?

Compare

Feature	Walking	Jumping
Peak amplitude		
Peak spacing		
Periodicity		
Detection valid?		

A model does not fail randomly. It fails when its assumptions are violated.

Final Reflection

- What is a “step” in your model?, What assumptions define it?, When does your system fail?
- Designing a wearable means defining what counts as a meaningful event.

Key Takeaway

Measure signals → apply models → infer meaning